

UNIVERSAL DOCKING PORT SYSTEM FOR INTERCONNECTING MODULAR CONTAINER-BASED STRUCTURES

Inventor: Chad Pattengale

[YOUR CITY], California

FIELD OF THE INVENTION

This invention relates generally to modular, relocatable structures built from or housed within standard intermodal shipping containers, and more particularly to systems and apparatus for mechanically and functionally interconnecting two or more separate container-based structures so that utilities such as electrical power, data and network connectivity, ventilation and conditioned air, and potable water may be shared between them, while also permitting each structure to operate independently using local standalone sources when not interconnected.

BACKGROUND OF THE INVENTION

Modular structures built from or incorporating standard 40-foot High Cube (40HC) intermodal shipping containers are increasingly used for rapid, relocatable construction, including for disaster-response and other applications where conventional building infrastructure is unavailable, damaged, or impractical to use. Existing approaches to interconnecting separate container-based units for shared utilities typically require permanent, custom, site-specific modification of the container structure; are not standardized across utility types; do not provide a reliable mechanical means of structurally aligning and securing adjacent units without external

power; or do not provide each unit with the ability to operate in a standalone mode using local utility sources when interconnection is unavailable or undesired.

There remains a need for a standardized, reusable docking interface that can be installed into the existing factory-built door-frame opening of a standard shipping container, without requiring a separate dedicated opening to be cut into the container structure, and that addresses multiple utility types and structural alignment through a single integrated mechanism suitable for rapid deployment by personnel without specialized training.

SUMMARY OF THE INVENTION

In one aspect, the invention provides a docking port bulkhead structure (100) sized and configured to be installed within the standard factory door-frame opening of a 40-foot High Cube intermodal shipping container in place of the container's original swinging cargo doors, while reusing the container's existing door hardware mounting points. The bulkhead structure is provided in at least two complementary variants: a first variant, associated with a utility-supply role, carrying male-type alignment and utility-zone connectors; and a second variant, associated with a demand role, carrying complementary female-type alignment and utility-zone connectors. When the door-frame opening of a first container-based structure fitted with the first variant is brought into facing alignment with the door-frame opening of a second container-based structure fitted with the second variant, the two bulkhead structures may be mechanically aligned, structurally locked together without external power, and the respective utility-zone connectors of each structure brought into engagement, establishing shared electrical, data, ventilation, and potable-water connections between the two structures, together with an integrated walk-through passageway permitting personnel and supplemental air movement between them.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are submitted herewith, illustrate representative views of the invention, including isometric views of two docking-port-equipped structures in both docked and undocked configurations; front elevation views of each bulkhead variant showing the layout of the structural alignment features, the passageway, and the four utility zones; and detail views of the structural locking mechanism, the electrical zone, the data zone, the ventilation zone, and the plumbing zone in both docked and standalone configurations.

DETAILED DESCRIPTION OF THE INVENTION

1. Overview and Mounting Architecture

The docking port bulkhead (100) is configured to install directly into the door-frame opening of a standard 40-foot High Cube (40HC) intermodal shipping container, which in a representative embodiment has nominal interior opening dimensions of approximately 2,340 mm in width and approximately 2,597 mm in height, in place of the container's original hinged cargo doors. In a representative embodiment, the bulkhead structure (100) has overall outer dimensions of approximately 2,200 mm in width, 2,500 mm in height, and 150 mm in depth, sized to be received within the container's factory door-frame opening with a perimeter clearance margin. The bulkhead is configured to mount to the container using the container's existing door hinge-ear and locking cam-retainer hardware, such that installation does not require cutting a new opening into the container's structural frame or end wall.

It will be understood that the specific dimensions given are illustrative of one embodiment sized for compatibility with a standard 40-foot High Cube container, and that the principles described herein may be adapted, without departing from the scope of the invention, to door-frame openings or bulkhead structures of other dimensions, including those associated with other standard or non-standard container sizes.

The bulkhead is provided in two structurally distinct variants rather than a single universal frame: a first, utility-supply-side variant intended for installation on a utility-supply unit, and a second, demand-side variant intended for installation on a demand unit. In the representative embodiment, the utility-supply-side variant carries the male half of the utility-zone connectors described below across all four utility zones, together with a pair of diagonally-opposed structural alignment pins (110); the demand-side variant carries the female half of each utility-zone connector, together with a pair of diagonally-opposed structural alignment pockets (120) positioned to receive the alignment pins (110) of a mated utility-supply-side unit. Providing two distinct, non-interchangeable frame variants, rather than a single bidirectional frame, reduces the risk of installing an incorrectly-configured unit in time-sensitive deployment scenarios, though in alternative embodiments a single bidirectional or reconfigurable frame carrying both male and female connector elements could be used instead.

2. Structural Alignment and Zero-Power Locking System

Each alignment pin (110) is a rigid cylindrical structural member, in a representative embodiment approximately 50 mm in diameter, configured to project forward from the bulkhead face by approximately 120 mm, and provided with a lead-in chamfer at its forward tip, in a representative embodiment at an angle of approximately 15 degrees from the pin's longitudinal axis, to guide and ease engagement with a corresponding alignment pocket (120) during docking.

Each corresponding alignment pocket (120) is a cylindrical recess, in a representative embodiment approximately 51.5 mm in internal diameter and approximately 125 mm in depth, sized to receive an alignment pin (110) with sufficient internal clearance to accommodate manufacturing and installation tolerances while preventing the pin from structurally bottoming out against the rear of the pocket.

Associated with the alignment pin and pocket structure is a mechanical locking mechanism (111) that engages automatically, without any electrical, hydraulic, or other powered actuation, as the

alignment pin is inserted into the alignment pocket, and that requires a deliberate manual release action (for example, operation of a lever, cam, or turnbuckle-type mechanism) to disengage. The locking mechanism is mechanically biased toward the engaged, locked configuration, for example by a spring or equivalent biasing element, such that the locking action occurs automatically upon insertion without requiring a separate locking step.

Because the locking mechanism requires no external power source to engage, hold, or be installed, it remains operable in environments where electrical power is unavailable or unreliable, which is of particular value in disaster-response and other infrastructure-degraded deployment scenarios. In alternative embodiments, the specific locking mechanism may be implemented using any mechanically-actuated, spring-biased, manually-releasable locking arrangement, including but not limited to cam-actuated, turnbuckle-actuated, or detent or latch-type mechanisms, and is not limited to any one specific mechanical implementation, so long as the engagement is automatic on insertion and release requires a deliberate manual action.

3. Passageway

Each bulkhead variant additionally incorporates a walk-through passageway opening, in a representative embodiment approximately 711 mm (28 inches) in clear width and extending substantially the full height of the bulkhead frame, fitted with a door (130) that is mechanically linked (131), through the same engagement travel that actuates the structural alignment pin and locking mechanism described above, such that the passageway door opens automatically as two units are brought into the docked configuration and closes automatically, sealing against a perimeter gasket, as the units are undocked or separated.

Because this mechanical linkage is identical and direction-agnostic on both bulkhead variants, the passageway door function does not depend on which variant is installed on a given structure. When two units are docked, the open passageway provides both a route for personnel movement between the two structures and a supplemental bulk-airflow path that augments the dedicated

ventilation ducting described below, since the open cross-sectional area of the passageway substantially exceeds that of the dedicated ventilation ducts.

4. Zone Layout

In the representative embodiment, the bulkhead frame is divided into a header region extending across the upper portion of the frame and a main body region below it. The header region carries, from one side to the other, a first ventilation duct connector, a data connector, and a second ventilation duct connector. The main body region carries the passageway described above at its center, flanked on one side by an electrical utility zone and on the other side by a plumbing utility zone, each extending substantially the full height of the main body region below the header. The structural alignment pins or pockets are positioned at diagonally opposed corners of the frame.

It will be understood that this layout is one representative arrangement and that the relative positions of the four utility zones, the passageway, and the structural alignment features may be varied in alternative embodiments without departing from the scope of the invention.

5. Zone 1: Electrical

The electrical utility zone (200) provides a docked power path comprising a power connector block (210), in a representative embodiment configured to provide a three-phase alternating current connection at approximately 480 volts and approximately 100 amperes through a quick-connect, blind- or hot-mate-capable contact arrangement, feeding through a mechanical disconnect switch (211) and an associated gate-plate interlock (212), the gate-plate interlock being mechanically tied to the disconnect switch through a linkage rod (213) such that the power contacts are physically inaccessible unless the disconnect switch is in the open or off position, with the linkage incorporating a gravity-actuated fail-safe, to a pass-through electrical bus (217).

The pass-through bus (217) is configured to carry docked-mode power onward to a further unit chained downstream of the present unit while bypassing the present unit's own local step-down transformer and panel, and additionally feeds a branch to the present unit's own step-down transformer (214), which converts the docked three-phase supply to a local single-phase supply, in a representative embodiment 120/240 volts, feeding an automatic transfer switch (216) and a local branch-circuit panel (215).

For standalone operation, a local generator feeds the same automatic transfer switch (216), which is configured such that the docked supply and the standalone generator supply are never both connected to the panel (215) simultaneously, and which automatically selects between the two sources depending on availability of the docked supply.

6. Zone 2: Data

The data utility zone (300) provides a docked data path comprising an all-optical-fiber connector (310), in a representative embodiment a multi-fiber cassette connector providing dual redundant optical signal paths over a single-mode fiber backbone, configured for blind-mate engagement using guide pins or guide holes, provided in complementary male and female form on the two bulkhead variants respectively, to ensure proper optical alignment without requiring the operator to visually align individual fibers during docking.

Because no electrically-conductive copper signal path crosses the docking interface in the data zone, the docked connection does not create a ground-loop path or lightning-surge exposure path between the electrical systems of the two docked structures. Each unit additionally incorporates a local network switch (311) coupled to the fiber connector (310) and providing local wired network ports for devices within that unit, together with a standalone data uplink (320), in a representative embodiment a satellite-based internet uplink module, coupled to the same local network switch (311) so that the unit retains network connectivity when operating in standalone mode without a docked connection.

A satellite-based standalone uplink is used in the representative embodiment because, in disaster-response deployment scenarios, terrestrial cellular and wireline data infrastructure are frequently among the first systems to be disrupted.

7. Zone 3: Ventilation

The ventilation utility zone (400) provides two round duct connectors, a supply duct connector (410) and a return duct connector (420), each in a representative embodiment approximately 305 mm (12 inches) in diameter. Each duct connector incorporates a mechanically-actuated auto-sealing damper (421), in a representative embodiment a butterfly-type damper disc, that is opened and closed through a cam-type linkage (422) driven by the same structural alignment-pin engagement travel described above, such that the damper opens automatically as the unit docks and closes automatically, sealing the duct, as the unit undocks, without requiring any electrical or other powered actuation.

Within each unit, an internal distribution manifold (430) receives air either from the docked duct supply, when docked, or from a local fan (431) and an associated local heat pump or air-conditioning unit (432), when operating in standalone mode, and distributes that air for further internal use. The local fan (431) is additionally configured to operate as a booster fan during docked pass-through operation, such that the fan (431) is used in both the docked and standalone operating modes, whereas the local heat pump or air-conditioning unit (432) is used only in the standalone operating mode, since conditioned air is supplied externally when docked. Where conditioned air passes through a unit's ducting to reach a further chained unit, the ducting may be insulated or wrapped to reduce thermal loss along that pass-through run.

8. Zone 4: Plumbing

The plumbing utility zone (500) provides potable water only; non-potable waste or discharge handling is provided separately and is not part of the bulkhead structure described herein. The

docked path comprises a no-drip, dry-break-type quick-disconnect fitting (510), feeding, in a representative embodiment, through a reduced-pressure-zone backflow-prevention device (520) configured to prevent backflow between the two structures' water systems.

A pass-through branch (530) is positioned upstream of the unit's own backflow-prevention device (520), such that potable water may be relayed onward to a further chained unit without being interrupted by servicing of the present unit's own backflow-prevention device or shutoff. For standalone operation, a local water storage tank or cistern, refillable from an external source such as a tanker truck, supplies the same internal distribution, the docked and standalone sources being selectable through a source-selection arrangement configured such that the two sources are not connected simultaneously.

9. Docked and Standalone Operating Modes

Each of the four utility zones described above is configured to operate in either of two modes: a docked mode, in which the zone's docked-side connector receives the relevant utility from a mated unit across the docking interface, or a standalone mode, in which the zone instead draws on a local source associated with that unit. In each zone, a switching arrangement is configured to prevent the docked-mode source and the standalone-mode source from being connected to the same load, panel, or manifold simultaneously.

10. Multi-Unit Chaining and Adapter Component

Because the utility-supply-side bulkhead variant carries male-type connectors and the demand-side bulkhead variant carries female-type connectors on both of its ends, a single utility-supply unit may dock directly to a demand unit at one of its two ends, leaving the demand unit's other, also female, end available to dock to a further unit.

Where it is desired to chain two demand units together end-to-end, each of which presents only female-type connectors, a separate, passive adapter component (600) is provided, carrying the

male half of all four utility-zone connectors, together with passageway and structural-alignment features matching the demand-side bulkhead geometry, on each of its two opposing faces, such that the adapter (600) may be coupled between the facing female ends of two demand units to bridge a docked connection between them. In the electrical and data zones, the adapter (600) provides a straight pass-through jumper without transformation; in the ventilation zone, it provides a duct coupling continuing the airflow path unmodified; and in the plumbing zone, it provides a pipe coupling continuing the potable-water pass-through line.

11. Alternative Embodiments

While the foregoing description refers to representative dimensions, materials, and specific component types for purposes of illustration, it will be understood by those skilled in the art that the invention is not limited to these specific examples, and that functionally equivalent alternative components, connector types, dimensions, materials, and mechanical arrangements may be substituted without departing from the scope of the invention.

Similarly, while the invention has been described in connection with a 40-foot High Cube intermodal shipping container, the underlying docking-port architecture — a door-frame-mounted bulkhead providing structural alignment, zero-power mechanical locking, multiple independently-switchable utility zones each supporting both docked and standalone operation, and an integrated passageway — may be adapted for use with other container sizes, other modular structure types, or other applications requiring a reusable, rapidly-deployable interconnection between separate relocatable structures.